

HEATWAVE & HEATSTROKE

From Thermoregulation to Emergency Management

MBBS Seminar · 45 Minutes

Learning Objectives

- 1 Explain physiology of thermoregulation and how it fails
- 2 Distinguish heat exhaustion from heat stroke (classic vs exertional)
- 3 Identify clinical red flags — especially CNS dysfunction
- 4 Initiate immediate cooling and supportive management
- 5 Counsel patients on heatwave prevention

How to Run This Session: Engaging Techniques

Open with a NEWS HEADLINE

Show an India heatwave death headline from May 2024. Ask: "What killed this person — and could you have saved them?" Hook first, teach second.

MYTH-BUSTING opener (Show of Hands)

"Raise your hand if you'd give paracetamol for heat stroke fever." Most raise hands. Then: "That's wrong — here's why." Students remember being corrected better than being taught.

"What would YOU do RIGHT NOW?"

Stop mid-case before revealing answers. Force commitment. Create cognitive conflict = better retention.

THE ANALOGY ANCHOR

Say once: "Heat stroke is sepsis — different trigger, same cytokine storm, same MOF logic." They already have the framework for every complication.

COOLING RACE demo

Draw 4 cooling methods. Ask students to rank by speed. Then reveal actual cooling rates. Counterintuitive = memorable.

Epidemiology — Why This Matters

40,000+

Suspected cases India 2024

May–June

Peak season

Preventable

Deaths are

High-Risk Groups

-  Elderly — impaired sweating
-  Infants — high surface-area ratio
-  Outdoor laborers, athletes, soldiers
-  On diuretics, anticholinergics, antipsychotics, beta-blockers
-  Chronic illness: DM, CKD, CVD, obesity, alcohol use



Global Warming Context

Heatwaves are increasing in BOTH frequency and intensity. IMD now issues colour-coded Heat Action alerts:

 Yellow: >40°C  Orange: >43°C  Red: >45°C

NAPHI (National Action Plan for Heat-related Illness) mandates district-level cooling centres.

Normal Thermoregulation – Quick Review

Set point: 37°C ($\pm 0.5^\circ\text{C}$)

Radiation

Heat emitted as infrared
(normal ambient temp)

Convection

Heat carried by
moving air

Conduction

Heat transferred
by direct contact

Evaporation (Sweating) ★

PRIMARY at $>35^\circ\text{C}$
abolished by humidity



KEY INSIGHT: When ambient temp $>$ skin temp, evaporation is the **ONLY** effective cooling mechanism.
High humidity $\rightarrow$ abolishes evaporation $\rightarrow$ thermoregulatory failure.

Hypothalamic integration: Anterior hypothalamus detects rise $\rightarrow$ Cutaneous vasodilation + sweating $\rightarrow$ Heat loss.
Above $41.5\text{--}42^\circ\text{C}$: Protein denaturation becomes CATASTROPHIC.

Pathophysiology of Heat Stroke

↑ Core Temp > 40°C

Protein denaturation
Cell membrane instability

Endothelial Injury

Cytokine storm
(IL-1, IL-6, TNF-α)

Systemic Inflammation

SIRS → MODS
Analogous to sepsis

Multi-Organ Failure

CNS · Liver · Kidney
Muscle · Coagulation

 **Critical Threshold: >40°C triggers irreversible cellular injury. Above 41.5°C = catastrophic protein denaturation.**

 **Key concept: Heat stroke is NOT just dehydration. It is a systemic inflammatory state — treat it like sepsis.**

This is almost always news to students — most think heat stroke = "too hot + no water."

Heat Illness Spectrum & Classification

Heat Cramps
(Mild)

Heat Syncope

Heat Exhaustion
(Moderate)

HEAT STROKE
(Emergency)

Feature	Heat Exhaustion	Heat Stroke
Core Temp	< 40°C	> 40°C ⚠
CNS Dysfunction	ABSENT	PRESENT (defining feature)
Sweating	Present	May be absent (classic)
Skin	Pale, clammy	Hot, flushed
Consciousness	Alert / dizzy	Confusion, seizure, coma
Mortality	< 1%	10–50% if untreated



CNS dysfunction (confusion, seizure, coma) = **HEAT STROKE** until proven otherwise. Act immediately.

Classic vs Exertional Heat Stroke

CLASSIC (Non-Exertional)

Who: Elderly, chronically ill

Onset: Gradual (days of heat exposure)

Sweating: Usually ABSENT

Rhabdomyolysis: Rare

DIC: Mild

CK levels: Mildly elevated

Example: "Elderly in a hot room during heatwave"

EXERTIONAL

Who: Young athletes, soldiers, laborers

Onset: Rapid (hours of intense activity)

Sweating: Often PRESENT

Rhabdomyolysis: COMMON ⚠️

DIC: SEVERE ⚠️

CK levels: MARKEDLY elevated

Example: "Marathon runner collapsing mid-race"

Clinical Features — Head to Toe

CNS (KEY)

Confusion, agitation, delirium → Seizures → Coma
Cerebellar ataxia — early and sensitive (often missed!)
Pupils may be dilated

Cardiovascular

Tachycardia (HR 140–170)
Initially hyperdynamic → high-output state
Hypotension as dehydration + vasodilation progress

Skin

Hot, DRY (classic) OR Hot, WET (exertional)
Flushed appearance

GI / Renal

Nausea, vomiting, diarrhoea (early gut ischaemia)
Oliguria → anuria (ATN from rhabdomyolysis)

Haematology

Petechiae, ecchymosis → DIC
Severe in exertional HS

Diagnosis — Clinical First, Labs After



Clinical diagnosis FIRST — do not wait for labs to start cooling.

- ① **Core temperature > 40°C** (rectal — axillary unreliable)
- ② **CNS dysfunction** (confusion / seizure / loss of consciousness)
- ③ History of heat exposure

Investigations — assess organ damage, not diagnose:

CBC: Hemoconcentration, thrombocytopenia (DIC)

PT/aPTT, D-dimer: DIC

RFT: AKI (↑ creatinine, urea)

ABG: Lactic acidosis, respiratory alkalosis

LFT: Hepatic injury — peaks at 48–72 h

Blood glucose: Hypoglycaemia (exertional)

CK / Myoglobin: Rhabdomyolysis

Electrolytes: Hypo/hypernatraemia

COOL FIRST, TRANSPORT SECOND

Every minute above 40°C = more organ damage

TARGET: Core temperature < 38.5°C within 30 minutes

STEP 1

Remove & Position

Move to cool area
Remove all clothing

STEP 2

Rapid Cooling

Begin immediately
(before IV access)

STEP 3

Supportive Care

Airway · Fluids
Seizure control

Cooling Methods — Ranked by Effectiveness

Ask students to rank these BEFORE showing the rates — counterintuitive answers = better retention



Cold Water Immersion
(0–15°C)

0.20°C / min

GOLD STANDARD

2nd

Ice Packs
(neck, axilla, groin)

0.10°C / min

Practical in field

3rd

Evaporative Cooling
(mist + fan)

0.10°C / min

Widely used in India

4th

Cool IV Fluids

Adjunct only

NOT primary cooling

⊖ Stop cooling at 38.5°C — risk of rebound hypothermia. ⊖ NO antipyretics (paracetamol / NSAIDs) — ineffective, may worsen AKI/liver.

Supportive Care

Airway

GCS $\leq$ 8 → Intubate
Protect airway

Seizures

IV Lorazepam 0.1 mg/kg
(or Diazepam 5–10 mg IV)
 Avoid Phenytoin

DIC

FFP + platelets per severity

Circulation

IV NS or RL
1–2 L bolus, titrate to UO 0.5–1 mL/kg/hr
Avoid fluid overload (cardiac patients)

Rhabdomyolysis

Aggressive IV fluids
Urine alkalinisation (IV NaHCO_3)
Target UO: 200–300 mL/hr

Hypoglycaemia

50% Dextrose IV

Monitor continuously: Rectal temp · Urine output · ECG · Electrolytes · Renal & liver function

⚡ WHAT STUDENTS ALMOST NEVER KNOW

CLINICAL BLINDSPOTS



Sweating does NOT rule out heat stroke — Exertional HS patients sweat heavily. Students assume "hot dry skin" is required. It is NOT.



Cerebellar ataxia = EARLY sign — Ataxia appears before coma. Students look for confusion and miss the wobbly gait.



Rectal temp is mandatory — Axillary temperature is unreliable at high core temps. Students routinely use axillary — which is wrong.



Hepatic injury peaks at 48–72 hours — Transaminases keep rising AFTER cooling. Students would discharge too early.



Heat intolerance persists for months — Hypothalamic dysregulation outlasts the acute episode. A surprise to almost everyone.

MANAGEMENT BLINDSPOTS

These are the highest-yield traps in exams AND clinical practice

 Paracetamol / NSAIDs

 NO — heat stroke has NO pyrogen. They won't work. NSAIDs also worsen AKI.

 Stop cooling at 37°C

 NO — stop at 38.5°C. Continuing causes rebound hypothermia.

 Phenytoin for seizures

 NO — ineffective in metabolic seizures + impairs heat dissipation. Use benzodiazepines.

 Furosemide for rhabdomyolysis

 NO — concentrates myoglobin in tubules, worsening AKI. Use aggressive IVF + NaHCO_3 .

 Transport first, cool later

 NO — "Cool first, transport second." Pre-hospital cooling saves lives.

Drug Interactions – High-Yield Blindspot

Students rarely connect medications to heat stroke risk. Ask: "What drugs predispose to heat stroke?"

Anticholinergics (Amitriptyline, Antipsychotics, Oxybutynin)

Block sweating → impairs evaporative cooling
→ Classic HS risk

Diuretics (Furosemide, Thiazides)

Volume depletion → reduces cardiovascular cooling reserve

Beta-blockers

Impair tachycardia response
Reduce cutaneous blood flow

Sympathomimetics (Cocaine, Amphetamines)

Increase heat production
+ cause vasoconstriction

Antipsychotics (Haloperidol, Chlorpromazine)

Impair hypothalamic thermoregulation
+ block sweating

Alcohol

Impairs hypothalamic function
+ causes dehydration

CASE 1

Classic Heat Stroke — Apply the Framework

Case 1 — Classic Heat Stroke

68-year-old male, retired. Found confused at home by neighbour on a May afternoon. History of hypertension on amlodipine + furosemide. No AC, windows closed. Daughter says he was "not himself" for past few hours.

Exam: GCS E3V3M5 · Temp 41.2°C (rectal) · BP 90/60 · HR 132 · Skin: Hot, DRY

Q1

What is the diagnosis? → Classic (non-exertional) Heat Stroke. Age + heat exposure + CNS dysfunction + temp >40°C + dry skin. Furosemide → volume depletion risk.

Q2

Immediate first action? → "Cool first, transport second." Strip clothes + ice packs to neck/axilla/groin + fan + misting. Do NOT wait for investigations.

Q3

Which medication contributed? → Furosemide (volume depletion) + amlodipine (impairs vasomotor response). Hold both. Replace volume cautiously — elderly cardiac reserve is limited.

CASE 2

Exertional Heat Stroke — Rhabdomyolysis Focus

Case 2 — Exertional Heat Stroke

24-year-old male, NCC cadet. Collapsed 3 km into a 5 km run. Humid 38°C day.
Brought by batchmates. Vomited once. Was talking initially, now confused.
Dark-coloured urine noted on catheterisation.

Exam: GCS E3V4M5 · Temp 40.6°C · BP 100/70 · HR 148 · Skin: Hot, SWEATING

Q1 **Classic or Exertional?** → EXERTIONAL. Young + exercise + sweating still present + dark urine = myoglobinuria → rhabdomyolysis.

Q2 **Dark urine — suspect? Manage?** → Rhabdomyolysis. Risk: tubular obstruction → AKI. Aggressive IVF (target UO 200–300 mL/hr) + IV NaHCO₃ (alkalinises urine, keeps myoglobin soluble). Monitor K⁺ (hyperkalemia risk).

Q3 **Should you give NSAIDs for fever?** → NO. Antipyretics don't work (no pyrogen). NSAIDs also worsen AKI and can cause GI bleeding.

Q4 **He seizes — what do you give?** → IV Lorazepam (0.1 mg/kg) or Diazepam (5–10 mg IV). NOT phenytoin — ineffective here + impairs heat dissipation.

MCQ ROUND

Test yourself — commit before revealing

MCQ Round — Questions 1 to 3

Q1. 70F, confused, temp 41°C, hot dry skin after 3-day heatwave. On amitriptyline. MOST important immediate step?

A) IV paracetamol 1g B) Lumbar puncture (rule out meningitis) C) Rapid external cooling ✓ D) IV morphine for agitation

C — Cooling is the priority. Amitriptyline (anticholinergic) blocks sweating → classic HS. LP can wait; cooling cannot.

Q2. Which BEST differentiates heat stroke from heat exhaustion?

A) Temperature > 38°C B) Tachycardia C) Presence of sweating D) CNS dysfunction ✓

D — CNS dysfunction is the defining hallmark of heat stroke. Temperature supports but CNS involvement differentiates.

Q3. 22M marathon runner, heat stroke, CK 45,000, dark urine. MOST important to prevent AKI?

A) Haemodialysis B) Aggressive IVF + urine alkalinisation ✓ C) IV furosemide to force diuresis D) Restrict fluids

B — Myoglobin is nephrotoxic in acidic urine. Alkalinisation keeps it soluble. Furosemide concentrates tubular myoglobin — avoid.

MCQ Round — Questions 4 to 5

Q4. Which cooling method has the FASTEST cooling rate for heat stroke?

A) Ice packs to axilla and groin

B) Evaporative cooling (mist + fan)

C) Cold water immersion (0–15°C) ✓

D) Cool IV fluids

Q5. A heat stroke patient develops a generalised tonic-clonic seizure. MOST appropriate anticonvulsant?

A) IV phenytoin

B) Oral carbamazepine

C) IV valproate

D) IV lorazepam ✓

► **D — Benzodiazepines are first-line. Phenytoin: (1) ineffective in metabolic seizures, (2) causes peripheral vasoconstriction → impairs heat dissipation.**

5 Things to Remember

- 1 Heat stroke = Core temp $>40^{\circ}\text{C}$ + CNS dysfunction — a true emergency
- 2 "Cool first, transport second" — every minute above 40°C costs organ function
- 3 Cold water immersion is the most effective cooling ($\sim 0.20^{\circ}\text{C}/\text{min}$)
- 4 NO antipyretics, NO phenytoin for seizures
- 5 Exertional HS $\rightarrow$ Rhabdomyolysis $\rightarrow$ aggressive IVF + urine alkalinisation